

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.6	Gender	Male
Height (cm)	175.0	Weight (kg)	85.0
BMI	27.8	Product Info 2	A1
Occupation	Pension	Annual Income	\$400,000.00
Insurance Type	Life	Coverage Amount	\$500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

****Applicant Details:****

- * Age: 0.6
- * Height: 175.0 cm, Weight: 85.0 kg, BMI: 27.8 (Overweight)
- * Smoker: Yes
- * Occupation: Pension
- * Income: \$400,000.00
- * Previous Claims: 0, Family History: 0
- * Exercise: 0 (Infrequent/None), Alcohol: 3 (Moderate/Heavy)
- * Insurance Type: Life, Coverage Amount: \$500,000.00
- * Gender: Male

****Machine Learning Classifier Output:****

- * Predicted Risk Class: 1 (Lower is better)
- * Prediction Probability/Confidence: 55.5%
- * Calculated Annual Premium: \$10,000.00

1. Risk Assessment

The applicant presents a mixed risk profile with several notable factors:

- * ****Age:**** The reported age of 0.6 years is a critical and highly anomalous data point. If interpreted literally as an infant, it directly contradicts the stated income, occupation, and the general premise of a life insurance application for an adult. This requires immediate clarification.
- * ****BMI:**** At 27.8, the applicant is classified as overweight, approaching the obese category (BMI > 30). This is a moderate health risk factor.
- * ****Smoking Status:**** The applicant is a smoker (indicator 1), which is a significant and adverse health risk factor, typically leading to higher premiums and/or lower risk classifications.
- * ****Occupation:**** "Pension" is an ambiguous occupation. If it implies a retired individual receiving a pension, it conflicts severely with the reported age of 0.6 years. If it refers to a professional in the pension industry, it's an incomplete description.
- * ****Income & Coverage:**** An income of \$400,000 with a coverage request of \$500,000 results in a coverage-to-income ratio of 1.25x, which is well within acceptable limits and does not suggest over-insurance. High income generally correlates with lower mortality for professionals.
- * ****Previous Claims & Family History:**** Both indicators are 0, which is favorable and reduces risk.
- * ****Lifestyle:**** Low exercise (0) and moderate/heavy alcohol consumption (3, assuming standard units) contribute to an elevated lifestyle risk profile.

2. Comparative Analysis

The machine learning model has classified this application as Risk Class 1 with a premium of \$10,000, aligning perfectly with three historical approved cases (APP-0642B5, POL-59031, POL-46791) that exhibit 100% similarity.

This 100% similarity suggests that, based on the input features provided, the current application is numerically identical to these previously approved cases. The consistency in Risk Class and Premium calculation (\$10,000.00) indicates that the model is functioning predictably when presented with this specific set of input data.

However, this raises a crucial point regarding the internal consistency and validity of the input data itself, particularly the "age" and "occupation" fields. If the historical cases also contained an age of 0.6 and occupation "Pension", it implies either a systemic data interpretation rule by the ML model that normalizes or overrides these values, or a consistent data entry error across multiple applications that the model has learned to process as a low-risk profile.

The relatively low prediction confidence (55.5%) for a Risk Class 1, especially given 100% similarity to approved cases, may suggest that while the primary classification is consistent, the model might be encountering some internal flags or uncertainties with the presented feature set, perhaps due to the unusual nature of certain inputs.

3. Anomaly & Fraud Detection

Several significant anomalies are present in this application:

- * **Critical Age Anomaly:** The most striking anomaly is the applicant's age of 0.6 years. This is highly improbable for an individual applying for life insurance with a stated income of \$400,000 and an occupation (even ambiguous) of "Pension." This strongly suggests a data entry error (e.g., 60 years entered as 0.6, or 0.6 decades, or 0.6 months misinterpreted as years) or a severe data integrity issue.
- * **Occupation Ambiguity:** The occupation "Pension" is vague and, coupled with the age anomaly, becomes highly suspicious. It provides insufficient information to properly assess occupational risk.
- * **Inconsistency in Risk Factors vs. Predicted Class:** Despite being a smoker, overweight (BMI 27.8), and having moderate/heavy alcohol consumption with low exercise, the application is classified as Risk Class 1. While high income and a clean claims history are positive, these negative health and lifestyle factors typically lead to a higher risk classification. The low confidence score (55.5%) for Risk Class 1 may be an internal indicator of this underlying inconsistency that the model is struggling with.

There are no immediate indicators of standard fraud patterns such as unusually high coverage relative to income, or a sudden surge in claims history. However, the data integrity issues (especially age and occupation) could mask underlying misrepresentation or fraud if not verified.

4. Premium Recommendation

The calculated annual premium of \$10,000.00 is directly aligned with the Machine Learning Classifier's output and the historical precedent of 100% similar approved cases.

- * The model has assigned a Risk Class of 1, which typically corresponds to the lowest premium tier.
- * The identical premium across all similar historical cases strengthens the justification from a model consistency perspective.

However, the validity of this premium is contingent upon the accuracy of the input data. Given the severe

anomalies identified (primarily age and occupation), the \$10,000.00 premium, while consistently generated by the model, cannot be confidently recommended without data verification. If the age or occupation information is corrected, the resulting risk class and premium are highly likely to change.

5. Underwriting Decision Recommendation

Given the critical data inconsistencies, specifically the applicant's reported age of 0.6 years and the ambiguous occupation of "Pension," this application cannot proceed directly to approval.

****Recommendation: Refer for Manual Review****

****Action Points for Manual Review:****

1. ****Verify Applicant Age:**** This is paramount. Clarify the correct birth date and current age of the applicant. This correction will fundamentally alter the risk assessment.
2. ****Clarify Occupation:**** Obtain a precise description of the applicant's occupation and employment status.
3. ****Re-evaluate Risk:**** After correcting the age and occupation, re-run the application through the ML classifier or conduct a manual underwriting assessment to determine the appropriate risk class and premium.
4. ****Investigate ML Model's Interpretation:**** If possible, investigate how the ML model processed an age of 0.6 years and "Pension" occupation to arrive at a Risk Class 1, especially given other adverse factors. This could highlight potential weaknesses or assumptions in the model's feature engineering or training data.

Without resolving these fundamental data issues, proceeding with the current application would expose the company to significant underwriting risk.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.